

USER AND USER EXPERIENCE RESEARCH



User Experience Research

1. Discover Stage
2. Explore Stage
3. Test Stage
4. Listening Stage

User Research

1. Steps
2. Benefits
3. User Interviews
4. User Personas

Defining User Experience

How users perceive a product, such as whether a smartwatch is seen as **sleek or chunky**, and their emotional reaction to it, such as whether people have a **positive** experience when using it. (**Hornbæk and Hertzum, 2017**)

Hassenzahl's (2010) model of the user experience

- **Pragmatic:** how **simple, practical**, and obvious it is for the user to achieve their goals
- **Hedonic:** how **evocative and stimulating** the interaction is to users

The User Experience

How a product behaves and is used by people in the real world

- The way people **feel** about it and their **pleasure** and **satisfaction** when **using it, looking at it, holding it, and opening or closing it**
- “Every product that is used by someone has a **user experience**: **newspapers, ketchup bottles, reclining armchairs, cardigan sweaters.**” (Garrett, 2010)
- “All aspects of the end-user's interaction with the company, **its services, and its products.** (Nielsen and Norman, 2014), Cannot **design a user experience**—only can design *for* a **user experience**

USER EXPERIENCE RESEARCH

1. A User Experience (UX) Researcher studies real users to find their needs, goals, and problems. They use data to guide UI/UX design teams so that websites and apps are easy and helpful to use.
2. User Experience Research is the systematic process of studying users, their goals, needs, behaviors, activities, problems and context of use in order to inform the design and evaluation of interactive products.
3. Designers need to understand users and their context before deciding what the interactive product should do.

What does a UX researcher do?

1. Collaborating with designers and stakeholders to understand research needs
2. Defining research questions and selecting appropriate methods of data collection
3. Developing budgets and timelines for research projects
4. Recruiting participants for research studies
5. Conducting design research studies and analysing the data collected
6. Transforming your findings into easily understandable insights
7. Presenting your findings to designers, developers, and other stakeholders

Five basic steps that UX researchers

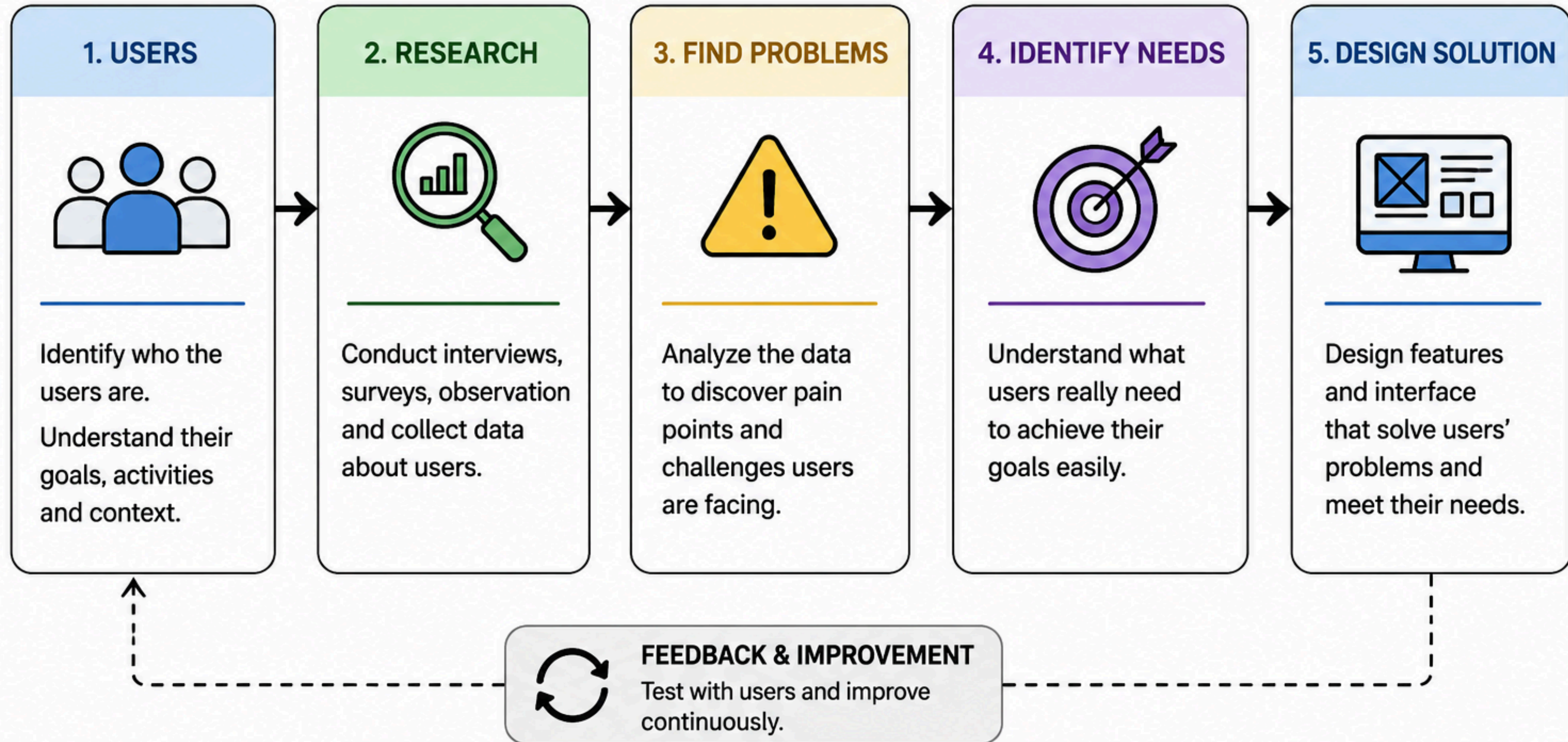
1. **Determine the objectives.** What is the average user looking for? How can you meet their needs?
2. **Develop your hypothesis.** What do you think the user is looking for? What niche is your product trying to fill?
3. **Create the methods.** Determine the scale of your research, how you are going to collect your data, and what research methods are most appropriate.
4. **Collect the data.** Follow the steps of your methods and make it a reality!
5. **Synthesise the data.** Now that you have collected the data, what are the findings? Do they fall in line with your expectations? If not, how do you need to change the product?

EXAMPLE

Suppose we are designing a College Attendance App. Instead of directly designing the screens, a UX researcher first talks to students and observes how they currently check their attendance. The research may reveal that students find it difficult to know which subjects have low attendance and often calculate the percentage manually. Based on these findings, the designer can create a dashboard showing subject-wise attendance, overall percentage, and shortage alerts.

Users → Research → Find Problems → Identify Needs → Design Solution

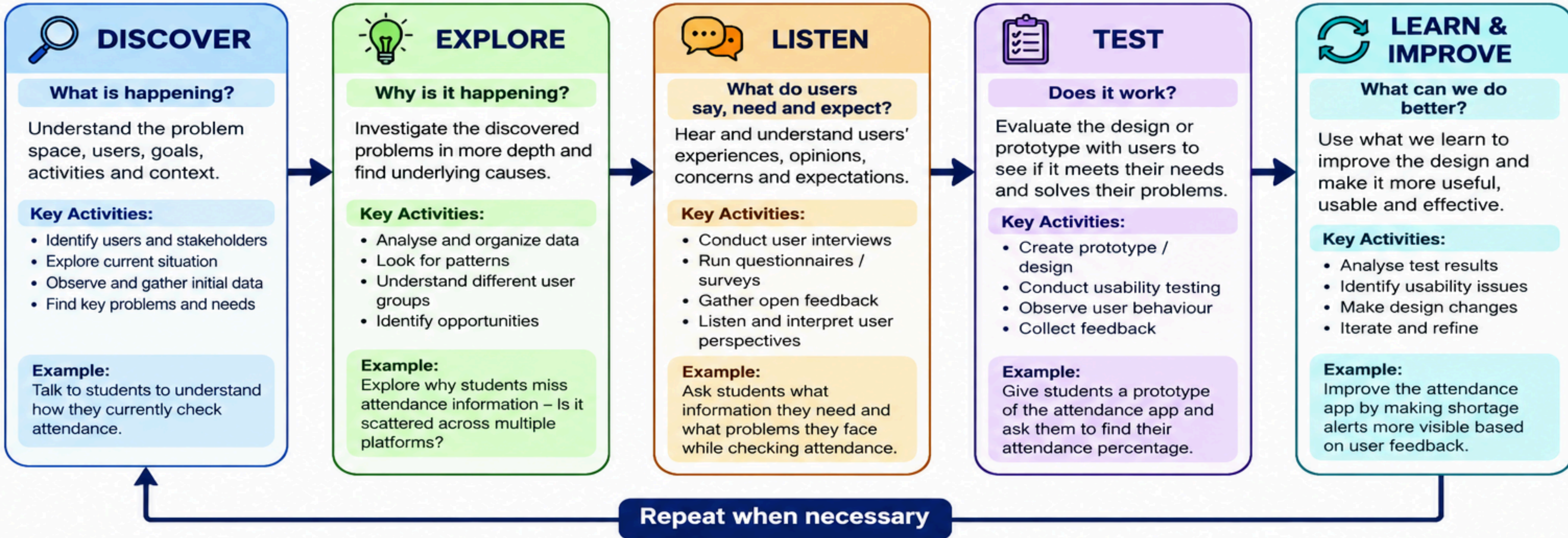
UX RESEARCH FLOW



Key Idea: Good design is not about the product, it's about solving **real user problems**.

USER EXPERIENCE RESEARCH — 4 STAGES

Understand → Explore → Listen → Test → Learn & Improve



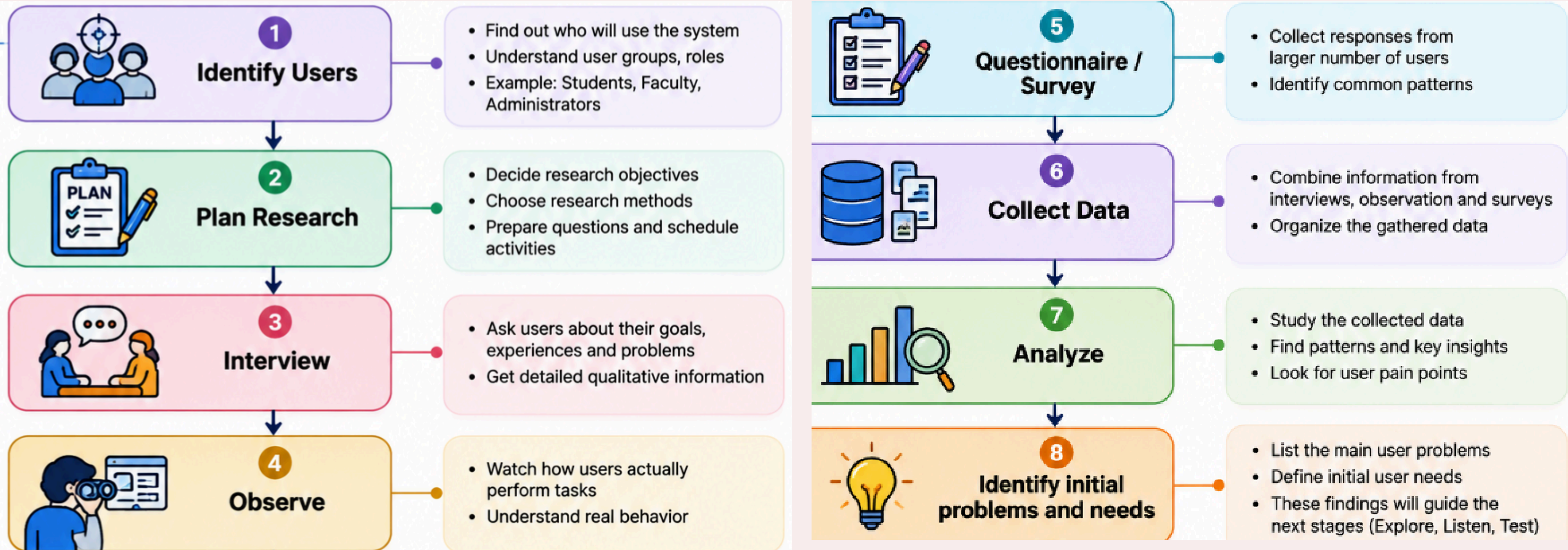
1. DISCOVER STAGE

- Discover means finding out who the users are, what they do, what they need and what problems exist before designing the solution.
- The Discover Stage is where we begin understanding the problem space, users, activities, goals and context before deciding on the design solution.
- They describe this activity as exploring the problem space and understanding potential users, their capabilities, current tasks, goals, contexts and constraints before defining what should be developed.

What do we discover?

1. Who are the users?
2. What are their goals?
3. What activities do they perform?
4. What problems do they face?
5. What is the context?

DESIGN PROCESS



How do we Discover?

1. Interviews
2. Questionnaires
3. Observation

1. Interviews

An interview is a direct conversation with a user where the researcher asks questions to understand the user's needs, problems, experiences, and expectations.

How it works

Researcher → Asks questions → User answers → Researcher records insights

2. Questionnaires

A questionnaire is a set of written questions given to many users to collect information in a structured and measurable way.

How it works

Researcher → Creates questions → Users respond → Responses are analyzed

3. Observation

Observation means watching users perform their real tasks to understand what they actually do, rather than only asking what they do.

How it works

Researcher → Watches user → Records behavior → Identifies problems

Technique	Main Purpose	Data	Best Question
Interview	Understand experiences and reasons	Mostly qualitative	Why? How?
Questionnaire	Collect opinions/preferences from many users	Quantitative + qualitative	How many? How often? How satisfied?
Observation	Understand actual behavior	Mostly behavioral/qualitative	What does the user actually do?

2. EXPLORE STAGE

- Explore means investigating the discovered problems, behaviors, needs and possible opportunities in greater detail.

Main purpose

- Investigate discovered problems.
- Understand the reasons behind user behavior.
- Explore user workflows.
- Identify patterns.
- Compare different user groups.
- Investigate pain points.
- Understand causes rather than just symptoms.
- Identify opportunities for improvement.
- Convert research observations into meaningful insights.

What happens in Explore?

1. Investigate the problem

Don't immediately assume the cause.

2. Ask deeper questions

Why does this happen?

3. Look for patterns

Do many users experience the same problem?

4. Understand different user groups

5. Identify opportunities

What could the system potentially improve?

Example – College Attendance App

Discover: Students say: "I don't know whether my attendance is sufficient."

Explore Researcher asks: "How do you currently find your attendance?"

Student: "I ask my class representative."

Another: "I check the college portal."

Another: "I ask the faculty."

Now we discover that there isn't one consistent way of accessing attendance information.

Attendance information is fragmented and not easily accessible.

Where should Explore be used?

- The problem is known but poorly understood.
- You need to understand causes.
- You need deeper user insights.
- Different users behave differently.
- You need to investigate workflows.
- You need to identify opportunities.
- You need to convert observations into meaningful requirements.

3. LISTENING STAGE

- The Listening Stage focuses on hearing and understanding users' experiences, opinions, concerns, expectations and needs.
- Listening means giving users the opportunity to explain their experiences and then carefully interpreting what they tell us.

Listening $\neq$ Just Hearing

- Ask $\rightarrow$ Listen $\rightarrow$ Clarify $\rightarrow$ Probe $\rightarrow$ Understand $\rightarrow$ Interpret

Main purpose:

- Understand user opinions.
- Understand user experiences.
- Discover expectations.
- Identify frustrations.
- Understand motivations.
- Understand emotional responses.
- Discover needs that users can articulate.
- Clarify unclear statements.
- Give users an opportunity to explain their experiences.

Techniques for Listening

1. User Interviews

Ask users questions.

Example:

"How do you currently check your attendance?"

2. Questionnaires

Ask the same questions to many users.

Example:

"How frequently do you check your attendance?"

- Daily
- Weekly
- Only before exams
- Other

3. Open-ended questions

Example:

"Tell me about the last time you checked your attendance."

This can produce richer information.

4. Follow-up questions

User:

"The portal is difficult."

Researcher:

"Can you tell me which part is difficult?"

Listening – What Are We Looking For?

- **Goals:** "I want to know my attendance quickly."
- **Frustrations:** "I have to open too many pages."
- **Expectations:** "I expected the percentage to be displayed on the dashboard."
- **Behaviors:** "I usually ask my friends."
- **Needs:** "I need a warning before my attendance becomes too low."
- **Emotions:** "I feel worried when I don't know whether I'm eligible."

Where should Listening be used?

- You need users' opinions.
- You need to understand experiences.
- You need to understand motivations.
- You need to discover frustrations.
- You need detailed qualitative information.
- You need to clarify something found during observation.
- You want to understand why users behave in a particular way.

4. TEST STAGE

- Testing means evaluating a design or prototype with users to identify problems and determine how well it supports their goals and tasks.

Main purpose

- Check whether users understand the design.
- Check whether users can complete tasks.
- Identify usability problems.
- Identify navigation problems.
- Check whether information is understandable.
- Evaluate effectiveness.
- Evaluate efficiency.
- Identify errors.
- Collect user feedback.
- Improve the design.

What is the Test Stage?

Research findings



User needs



Requirements



Design



Prototype

1. RESEARCH FINDINGS

- Research findings are the important facts, observations, patterns, and problems discovered from studying real users.

2. USER NEEDS

- User needs are the things users require or expect in order to achieve their goals effectively and conveniently.

3. REQUIREMENTS

- Requirements describe what the interactive system should provide or do to satisfy identified user needs.

4. DESIGN

- Design is the process of deciding how the product will look, behave and support users in completing their tasks based on the identified requirements.

5. PROTOTYPE

- A prototype is an early representation or working model of a design that allows designers and users to explore and evaluate how the proposed product will work before full implementation.

Example : Attendance App

Suppose we create this prototype:

Dashboard

- Overall Attendance: 82%
- Data Structures: 65%
- DBMS: 91%
- Mathematics: 78%

We now give the user a task:

"Find the subject where you have the lowest attendance."

Don't explain where to click.

Observe the user.

What do we observe?

1. Can they find it?
2. How long does it take?
3. Do they make errors?
4. Do they understand the information?
5. Do they know what to do next?
6. Are they confused?
7. Can they complete the task?

Where should Test be used?

Early

With sketches and low-fidelity prototypes.

During design

With wireframes and interactive prototypes.

Before implementation

To identify major usability problems.

After implementation

To evaluate the real product.

After changes

To check whether improvements actually worked.

1 DISCOVER STAGE

Understand the problem space, users, their goals, activities and context.



What we do

- Identify users
- Understand their goals
- Observe current behavior
- Identify initial problems
- Understand context

Example (College Attendance App)

We talk to students and observe how they currently check attendance. We discover that they use different sources (portal, WhatsApp, friends).



When to use

When starting a new project or entering an unknown domain. When we need to know who the users are and what problems exist.

2 EXPLORE STAGE

Go deeper to understand the problems, behaviors, causes and opportunities.



What we do

- Investigate the problem in detail
- Ask "Why?" and "How?"
- Look for patterns
- Study different user groups
- Identify opportunities

Example (College Attendance App)

We explore why students struggle. We find that information is scattered, attendance is hidden in multiple pages, and percentage is calculated manually.



When to use

When the problem is known but not clearly understood. When we need to find the root causes and deeper insights.

3 LISTENING STAGE

Listen to users to understand their experiences, needs, expectations and pain points.



What we do

- Conduct interviews
- Use questionnaires
- Ask open-ended questions
- Understand feelings, frustrations, needs and expectations

Example (College Attendance App)

Students tell us they need quick access, subject-wise attendance, shortage alerts, and a simple way to know if they are eligible for exams.



When to use

When we want to understand user opinions, expectations, motivations and detailed experiences.

4 TEST STAGE

Evaluate the design/prototype with users to see if it works and solves the problem.



What we do

- Create prototype
- Give tasks to users
- Observe behavior
- Measure task success & errors
- Collect feedback

Example (College Attendance App)

We give the prototype to students and ask them to find the subject with the lowest attendance. We observe where they get confused and what is not clear.



When to use

When we have a design or prototype and want to evaluate its usability and effectiveness before/after development.

DIFFERENCES BETWEEN THE 4 STAGES

Aspect	DISCOVER	EXPLORE	LISTEN	TEST
Main Question	What is happening?	Why is it happening?	What do users say, need and expect?	Does our solution work?
Purpose	Understand the broad picture of users, activities, context and initial problems.	Go deeper to understand causes, patterns and opportunities.	Understand user experiences, opinions, needs and expectations.	Evaluate design/prototype with real users to identify issues and validate.
Type of Data	General, broad, exploratory data	Detailed, investigative data	Qualitative (opinions, feelings, needs, expectations)	Performance data, behavior data, feedback data
Methods	Interviews, Observation, Surveys, Contextual inquiry	Task analysis, Workflow analysis, Observation, Interviews	Interviews, Questionnaires, Focus groups, Open questions	Usability testing, A/B testing, Prototype testing, Surveys
Output	List of users, goals, activities, initial problems, context	Deeper insights, patterns, root causes, opportunities	User needs, expectations, pain points, quotes	Usability issues, task success rate, feedback, improvement areas
When to Use	At the beginning of a project	After discovering initial problems	When we want to know what users think, feel and need	When we have a design/prototype to evaluate

VISUAL FLOW OF THE 4 STAGES



1.User Research

- User Research is the process of studying users to understand their needs, goals, behaviors, problems, and expectations before designing a product.

1. Steps of User Research

- 1. Identify the Problem**
- 2. Identify Target Users**
- 3. Plan Research**
- 4. Collect Data**
- 5. Analyze Data**
- 6. Identify User Needs**
- 7. Create User Personas**
- 8. Define Requirements**
- 9. Design the Solution**

Steps of User Research

1. College Bus Tracking App - Recommended

Scenario:

Students often wait for the college bus without knowing its current location or expected arrival time. Sometimes they miss the bus because of delays or changes in the route.

Goal:

Design a College Bus Tracking App that allows students to view the bus location, estimated arrival time, route, and notifications about delays.

Research questions:

How do students currently know when the bus will arrive?

How long do they usually wait?

What happens when the bus is delayed?

Do students want live bus tracking?

What information is most important to them?

Problem

→ Students don't know where the bus is.

Target Users

→ Students, bus drivers, transport coordinator.

Research

→ Interview + Questionnaire + Observation.

Findings

→ Students wait too long, don't know delays, etc.

User Needs

→ Live location + ETA + delay notification.

Persona

→ Student who travels by college bus.

Requirements

→ Track bus, show ETA, notifications.

Design

→ Home → Select Bus → Live Location → ETA → Notification.

Benefits of User Research

1. Understand Users

We understand who the users are and what they need.

2. Identify Real Problems

Instead of assuming the problem, we discover the actual user problem.

3. Reduce Assumptions

Without research:

"I think users will like this."

With research:

"Users told us they need this."

4. Improve Usability

Research helps us create products that are easier to use.

3. User Interviews

A User Interview is a direct conversation with users to understand their experiences, needs, motivations, problems, and expectations.

User says → Experience → Problem → Need

Purpose of User Interview

- What problems users face
- What users need
- What users expect
- How users currently perform a task
- What frustrates users
- Why users behave in a particular way
- What improvements users want

Interview = Talk to users to understand their WHY, HOW, and EXPERIENCE

Scenario: IT Company Employee Helpdesk System

Problem Scenario

Employees in an IT company face technical problems such as password issues, software installation, network problems, laptop problems, and access requests. Currently, employees may send emails, messages, or call the IT support team. They don't always know whether their issue has been received, who is handling it, or when it will be resolved.

Goal

Design an IT Helpdesk System that allows employees to report technical problems, track ticket status, and communicate with the IT support team.

4. User Personas

- A User Persona is a fictional character created from real user research to represent a typical target user.

It helps designers answer:

"Who are we designing this product for?"

- A User Persona is a fictional but research-based representation of a typical target user. It helps the design team understand who they are designing for.

Purpose of User Personas

- 1. Understand the Target User**
- 2. Understand User Goals**
- 3. Identify User Problems**
- 4. Identify User Needs**
- 5. Guide Design Decisions**
- 6. Maintain User Focus**
- 7. Create Empathy**
- 8. Prioritize Features**
- 9. Communicate with the Team**

What Does a Persona Usually Contain?

- **Name**
- **Age**
- **Role / Occupation**
- **Goals**
- **Problems / Pain Points**
- **Needs**
- **Behaviors**
- **Technology Usage**
- **Frustrations**
- **Motivations**

Scenario: Employee Leave Management System

Problem

In an IT company, employees currently request leave through email or messages to their manager. Employees sometimes don't know whether their leave has been approved, rejected, or is still pending.

Goal

Design an Employee Leave Management System where employees can apply for leave, check their leave balance, and track approval status

Topic	EXPLANATION
User Research	Studying users to understand their needs, goals, behaviors, and problems before designing a product.
Steps	Identify problem → Identify users → Plan → Collect data → Analyze → Identify needs → Create personas →
Benefits	User research helps identify real problems, reduce assumptions, improve usability, and make better design
User Interview	A direct conversation with users to understand their experiences, needs, motivations, and problems.
User Persona	A research-based fictional representation of a typical target user.

Design Tools & Methods



Design Tools

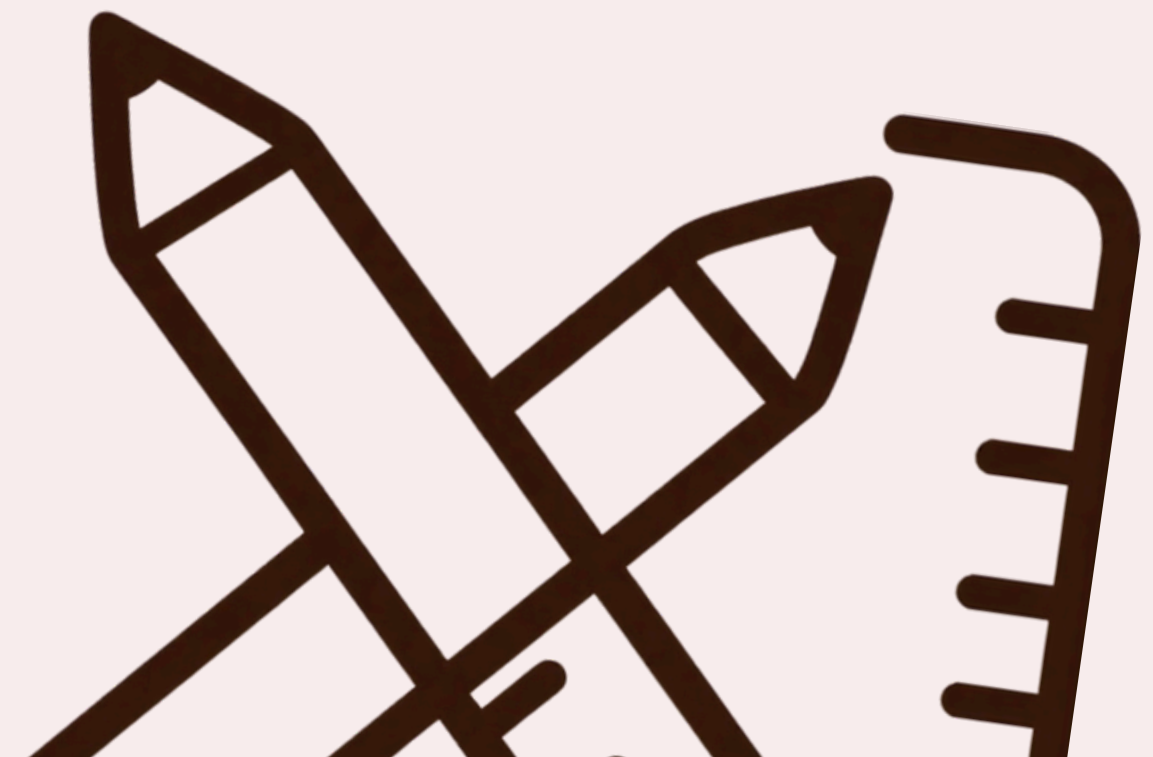
Figma, Adobe XD, and Sketch are industry-standard tools for UI/UX design. They enable designers to create layouts, components, and interactive prototypes efficiently.

Wireframing

Wireframes are low-fidelity blueprints of a screen. They define layout structure, content hierarchy, and user flow before visual design begins.

Prototyping

Prototypes simulate real interactions. From clickable mockups to high-fidelity demos, they help validate design decisions and gather user feedback early.



**GREAT DESIGN
IS INVISIBLE**

